RESEARCH MILESTONES

EMPATHIZE

Exploring Turkish tea culture, hospitality habits, and the challenges of controlling tea strength and quantity during preparation.

DEFINE

Identifying the main problem: difficulty in controlling tea strength and the waste caused by hospitality-driven over-preparation.

IDEATE

Developing concepts that translate the traditional Turkish tea brewing process into a controllable and automated system.

PROTOTYPE

Transforming the concept into a compact tea machine with a controlled brewing mechanism and adjustable tea strength.

TEST

Testing the clarity and usability of the interface for selecting the number of people and adjusting tea strength.

MILESTONE 1 : EMPATHIZE

Tea ritual • Sensory experience • Social interaction

Promt

According to these interview questions which questions would be most efficient for an interview, choose or add if needed.'

AI Response Summary

The response refined the interview questions to focus on emotional motivations, sensory experiences, and behavioral patterns during beverage preparation.

AI Role

Clarifier

Reflection

This helped frame beverage preparation not just as a task but as a moment of emotional reset and social interaction.

Accepted

- Emotional motivations, sensory elements (smell, steam, sound), and environmental context.

Rejected

- Purely technical or efficiency-based questions
- Questions that focused only on device performance rather than user experience

Transformed

- The interview shifted from a functional questionnaire to an experiential exploration of preparation rituals.

Promt

Considering the observations, interpretations, and insights derived from the Turkish coffee interview, explore how similar observation–interpretation–insight relationships might appear in the context of Turkish tea preparation. Identify potential tensions within the tea ritual and define opportunity areas for design intervention.

AI Response Summary

The response applied the observation–interpretation–insight framework to Turkish tea preparation, highlighting tensions between manual control, brew strength, and serving multiple guests.

AI Role

Expander

Reflection

This showed that Turkish tea preparation involves continuous adjustment rather than a single brewing moment.

Accepted

- Tea as a cultural ritual, visible brewing transformation, and multi-person serving.

Rejected

- Fully automated systems and hidden brewing processes.

Transformed

- The ritual was reframed as a dynamic system of brewing, dilution, and serving adjustments.

MILESTONE 2 : DEFINE

Hospitality culture • Portion uncertainty • Waste awareness

Promt

Analyze how Turkish tea became widely popular in Turkey during the 1980s and how it evolved from a common beverage into a deeply rooted cultural ritual. Please address:

• Economic and political factors
• Agricultural production and accessibility
• Domestic consumption patterns
• Hospitality and social rituals
• How tea shifted from a drink to a cultural symbol

AI Response Summary

The response explained tea's strong cultural role in Turkey and how hospitality norms encourage preparing tea in abundance.

AI Role

Expander

Reflection

This revealed a tension between cultural generosity and issues like waste, freshness, and portion control.

Accepted

- Tea as a symbol of hospitality and the norm of preparing extra tea.

Rejected

- Viewing tea preparation only as a functional process.

Transformed

- Tea preparation was reframed as a culturally influenced system, not just a technical brewing process.

Promt Summary

Based on the identified problem of uncertainty in tea and water ratios during Turkish tea preparation, define the most suitable primary user profile. The user should reflect tea rituals, fluctuating guest numbers, and sensitivity to waste.

AI Response Summary

The response defined a primary user: a 25–35-year-old urban individual who drinks tea daily and sometimes hosts small gatherings, often struggling to adjust tea quantities.

AI Role

Expander

Reflection

This showed that the problem occurs frequently in everyday life, especially when guest numbers change.

Accepted

- Socially active urban users who regularly prepare tea at home.

Rejected

- Users who prepare tea rarely
- Large-scale serving contexts such as cafés or restaurants

Transformed

- The general concept of "tea drinkers" was refined into a specific lifestyle-based user profile that regularly experiences the tension between tea preparation and portion control.

MILESTONE 3 : IDEATE

Electronic guidance • Portion control • Preventive brewing

Promt Summary

Based on the tension between abundance-oriented hospitality and precise portion control in Turkish tea culture, generate three electronic-assisted concept variations. The concepts should preserve the traditional double-teapot structure and manual brewing ritual, avoiding full automation. Electronics should only provide subtle guidance to help users manage changing guest numbers and reduce over-preparation, freshness loss, and waste."

AI Response Summary

The response proposed three concepts: water quantity guidance, tea freshness awareness, and guest-based brewing guidance, using subtle electronic feedback instead of automation.

AI Role

Expander

Reflection

This showed that the key moment for preventing waste is the initial water filling stage.

Accepted

- Volume awareness during filling, subtle electronic feedback, and manual brewing control.

Rejected

- Concepts focused only on serving or freshness, and overly complex technological systems.

Transformed

- The concepts were combined into one system focused on preventive guidance during preparation.

Promt Summary

Based on the previously generated concept directions, develop a single integrated Turkish tea system that combines water calibration, freshness awareness, and guest-based portion guidance into one coherent electronic-assisted system.
The system should preserve the traditional brewing ritual and use electronics only as subtle awareness support.

AI Response Summary

The response combined the concepts into a unified system called "Measured Generosity," supporting tea preparation through guest selection, water-volume awareness, and subtle freshness feedback.

AI Role

Re-framer

Reflection

This showed that preventing over-preparation during the initial water-filling stage is more critical than later feedback.

Accepted

- Guest intention as a starting point and subtle ambient guidance.

Rejected

- Multi-stage guidance systems and feature-heavy solutions.

Transformed

- The concept evolved into a focused preventive system centered on the initial preparation stage, where water and tea leaves are added.

MILESTONE 4 : PROTOTYPE

Brewing capacity • Component access • System organization

Promt Summary

I am designing a smart Turkish tea kettle based on the traditional double-teapot system (çaydanlık). The concept aims to guide the user during preparation by visually indicating the correct water level using light signals. To define the correct capacity and water level indicators for the product, realistic brewing quantities used in traditional Turkish tea preparation need to be understood.
For example, if tea is prepared for approximately 6 people, how much water is typically used in.

AI Response Summary

AI provided typical brewing capacities for Turkish tea preparation.
For about 6 people:
• Upper teapot (demlik): ~600–720 ml brewing water
• Lower kettle: ~1400–1600 ml hot water

AI Role

Expander

Reflection

This information helped define realistic product capacities and supported the design of a water-level guidance system. However, the concept remained strongly tied to the traditional kettle structure and did not significantly rethink the preparation process.

Accepted

- Using realistic Turkish tea brewing quantities and defining kettle capacity based on typical multi-person preparation.

Rejected

- Arbitrary capacities and purely decorative kettle concepts without functional guidance.

Transformed

- The concept shifted from a general smart kettle idea to a design based on real tea brewing ratios.

Promt Summary

I am refining the second concept of my Turkish tea machine using the existing visual design. The overall form stays the same, but some components still need to be clarified, such as the removable water reservoir and the tea filter. Where should these components be placed and how should the filter be accessed for cleaning so the interaction remains clear and intuitive?

AI Response Summary

The response suggested organizing the internal components to make the interaction clear and accessible. It recommended placing the removable water reservoir on the side or back of the machine for easy refilling, positioning the tea leaves container in the top section, and designing the tea filter as a removable part that can be accessed from the front for cleaning.

AI Role

Re-framer

Reflection

This feedback helped clarify how the internal components could be organized without changing the overall form of the product. It particularly highlighted the importance of making the filter and water reservoir easy to access for everyday use and maintenance.

Accepted

- A removable water reservoir for easy refilling.
- A removable tea filter that can be accessed and cleaned easily.

Rejected

- Hidden components that would make cleaning or refilling difficult.

Transformed

- The existing concept was refined by clarifying the placement and accessibility of internal components while keeping the original visual design.

MILESTONE 5 : TEST

Interface usability • Interaction clarity • Strength control

Promt Summary

I designed a compact automatic Turkish tea machine for a single-person household. The interface is located above the tea dispensing area and shows the number of people and the tea strength.
The interface includes:
• cup icons representing 1–6 people
• tea strength options (Light / Medium / Strong)
Can you evaluate whether this interface layout would be clear and intuitive for the user? Are there any usability issues or improvements that could make the interaction easier to understand?

AI Response Summary

AI evaluated the interface and suggested that using clear icons for the number of people and a simple tea strength scale helps users understand the system quickly. The minimal interface also supports the goal of making the machine easy to use.

AI Role

Critique

Reflection

This showed that the key moment for preventing waste is the initial water filling stage.

Accepted

- Volume awareness during filling, subtle electronic feedback, and manual brewing control.

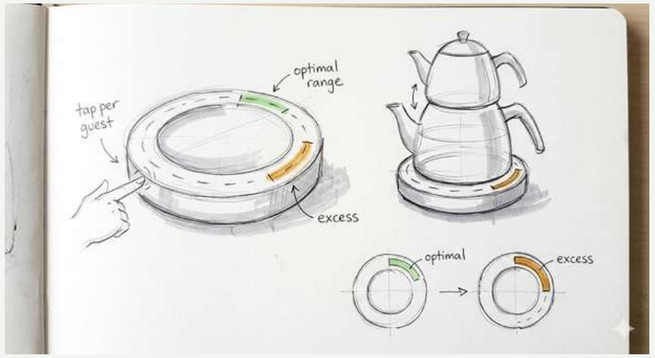
Rejected

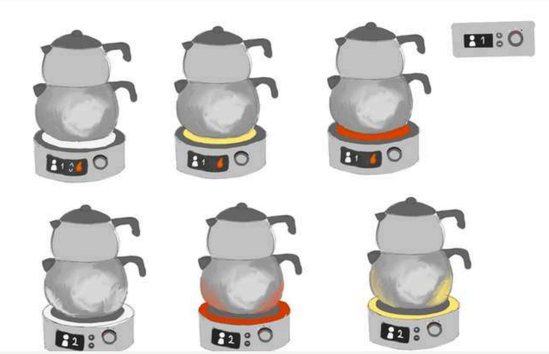
- Concepts focused only on serving or freshness, and overly complex technological systems.

Transformed

- The concepts were combined into one system focused on preventive guidance during preparation.

DESIGN DECISION: INITIAL





The design process initially started by following the traditional Turkish tea kettle form. The aim was to integrate an electronic system into the familiar double-teapot structure.

However, during this stage I realized that the traditional form was limiting the design possibilities. The structure of the teapot was controlling the system too much and preventing me from answering the main questions of the project. The concept was still highly dependent on the ritual and physical structure of the traditional tea kettle, which made it difficult to simplify the brewing process.

DESIGN DECISION: REFINEMENT

At this stage I began to rethink the system and approached the problem similar to a coffee machine logic.
Instead of focusing on the traditional structure, I started exploring the idea of a machine where:

- the user adds tea leaves and water
- selects the number of people
- and the machine prepares the appropriate amount of tea.

This direction helped answer the initial problem of portion control and reducing waste.

However, I still tried to preserve the traditional tea kettle form within the machine. As the design developed, the machine began to grow larger and more complex, gradually moving away from the intended user scenario of a single-person household.

The design was starting to resemble a machine more suitable for office environments.

DESIGN DECISION: FINAL

In the final stage I decided to completely move away from the traditional tea kettle structure and simplify the system.
The final concept focuses on a compact automated tea machine.
The interaction became much simpler:

1. The user adds tea leaves and water.
2. The user enters the number of people.
3. Once the tea is brewed, the brew strength can be adjusted for each glass individually.

This approach removes the constraints of the traditional form and creates a smaller, simpler, and more user-focused machine designed for everyday home use.